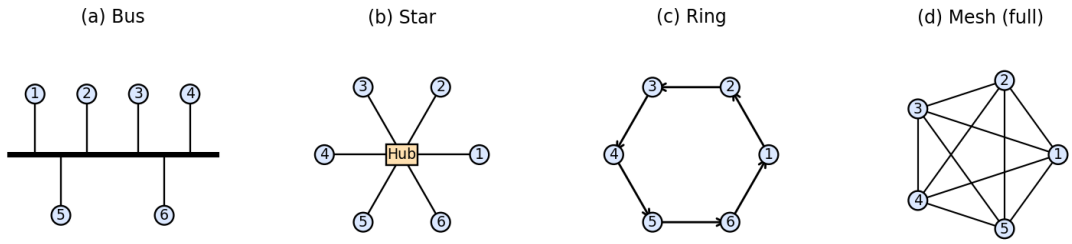
MCS-218: Data Communication and Computer Networks

Assignment MCA_NEW/MCAOL(II)/218/Assign/2026-27 (Simple Answers)

Name: _____ Enrolment No.: _____

Q1. Network topologies



Bus, Star, Ring, Mesh

Topology	How it works	Advantages	Disadvantages
Bus	All devices share one main cable.	Cheap, easy, less cable	Cable break stops everything; slow when busy
Star	All devices connect to a central hub/switch.	Easy to add devices; one link failure affects only one device	Hub failure stops everything
Ring	Devices form a closed loop; data goes round it.	Orderly, no collisions	One break can stop the ring; hard to add devices
Mesh	Every device is linked to every other.	Very reliable, many paths	Very costly, too many cables

Q2. Bandwidth

Bandwidth = highest frequency - lowest frequency = 7000 - 2000 = **5000 Hz (5 kHz)**.

Q3. Modulation techniques

Technique	Input	What changes	Use
FM	Analog	Carrier frequency changes with the signal	FM radio
PCM	Analog	Signal is sampled, quantized, then converted to binary codes	Digital telephony
FSK	Digital	Carrier frequency switches between two values for 0 and 1	Modems
PAM	Analog	Height (amplitude) of pulses changes with the signal	First step of PCM

Q4. Maximum amplitude levels

Range = 7 - (-5) = 12 V. Minimum gap = 0.6 V.

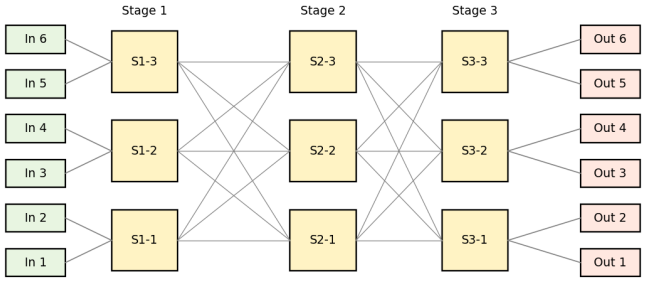
Gaps = 12 / 0.6 = 20, so levels = 20 + 1 = **21 levels**.

Q5. Switching

a. Significance: Switching joins many devices through shared links, so we do not need a direct cable between every pair. It saves cost and carries data from source to destination.

b. Techniques:

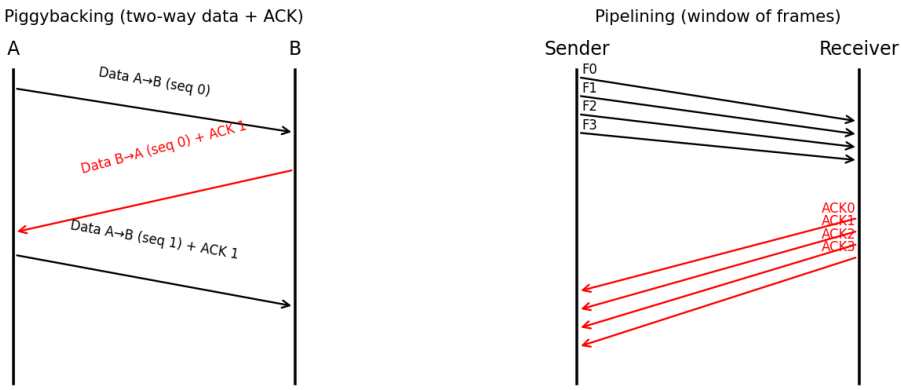
- **Circuit switching:** A fixed path is set up first (like a phone call), data is sent, then the path is released. Fast and steady, but wastes the line when idle.
- **Multistage switching:** Many small switches arranged in stages (input, middle, output) replace one big switch. It needs far fewer crosspoints. A free path is found through the middle stage.
- **Packet switching:** Data is cut into packets, each with an address. Packets travel separately and are joined at the destination. Efficient for bursty data, but delay can vary.



Multistage switch

Q6. Retransmission terms

- **Piggybacking:** The acknowledgement is attached to a data frame going the other way, instead of sending a separate ACK. This saves bandwidth.
- **Pipelining:** The sender sends many frames one after another without waiting for each ACK, so the line stays busy (used in Go-Back-N and Selective Repeat).

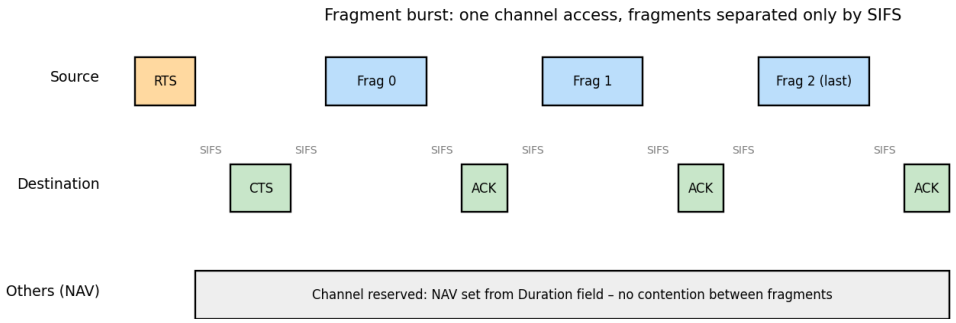


Piggybacking and pipelining

Q7. 802.11 and Ethernet

a. Fragment burst: A big frame is split into small fragments, because wireless errors are common. All fragments are sent in one go:

- Sender wins the channel (optionally RTS/CTS).
- Sends fragment 0; others stay silent (NAV set).
- Receiver sends ACK after a short gap (SIFS).
- Sender sends next fragment after SIFS, gets ACK, and repeats until the last fragment.



Fragment burst

b. Binary exponential backoff: After a collision, a station waits a random number of time slots before retrying.

- After the n-th collision, pick K randomly from 0 to 2^n - 1 (n limited to 10).
- Wait K slots (1 slot = 51.2 microseconds), then try again.
- After 16 failures, give up and report an error.

The waiting range doubles each time, so repeated collisions become less likely.

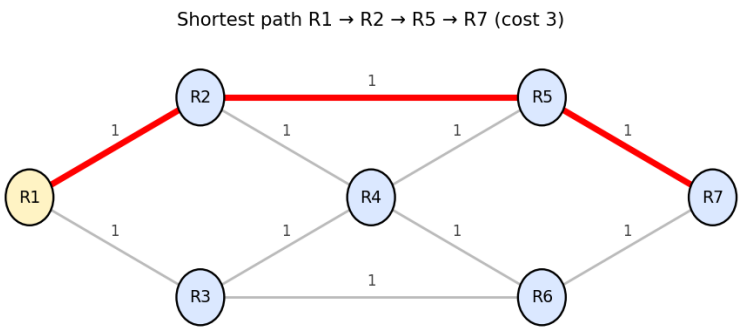
Q8. Routing and flooding

Assumption: Figure 1 does not show link costs (only the t= flood times), so each link cost is taken as 1.

a. Dijkstra from R1:

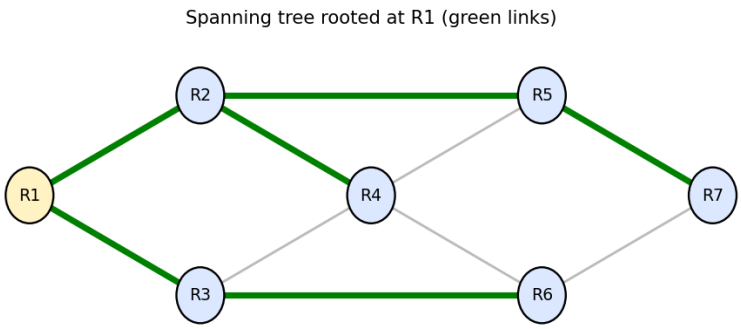
Step	Fixed node	R2	R3	R4	R5	R6	R7
1	R1	1	1	inf	inf	inf	inf
2	R2	1	1	2	2	inf	inf
3	R3	1	1	2	2	2	inf
4	R4			2	2	2	inf
5	R5				2	2	3 (via R5)
6	R6					2	3
7	R7						3

Shortest path: R1 - R2 - R5 - R7, cost = 3.



Shortest path

b. Spanning tree (root R1): links R1-R2, R1-R3, R2-R4, R2-R5, R3-R6, R5-R7 (6 links, no loops).



Spanning tree

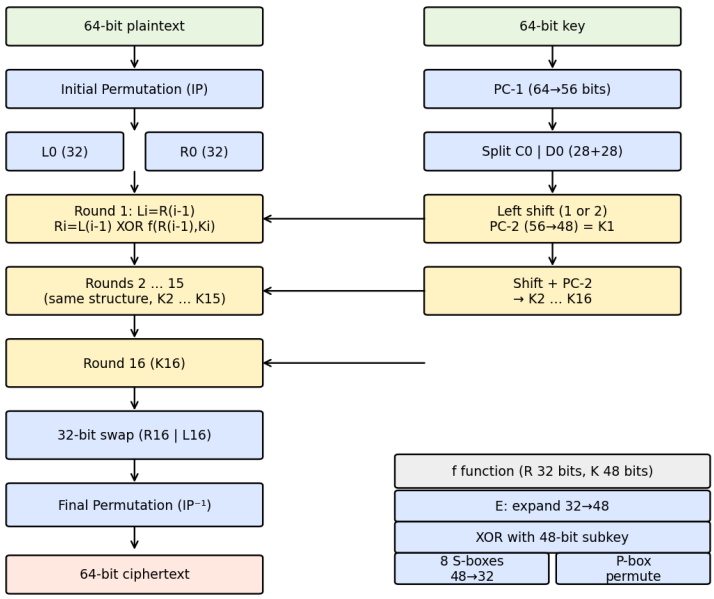
Flooding without tree: t=0: 2 packets; t=1: 4 packets; t=2: 6 packets. Total = **12** (6 are duplicates).

Flooding with tree: t=0: 2; t=1: 3; t=2: 1. Total = **6**.

So the spanning tree halves the traffic (12 to 6) and removes duplicates.

Q9. Data Encryption Standard (DES)

DES is a symmetric block cipher: the same key encrypts and decrypts. It takes a **64-bit block**, uses a **56-bit key** and has **16 rounds**.



DES architecture

Algorithm:

- Apply the Initial Permutation (IP) to the 64-bit block and split into left (L) and right (R) halves of 32 bits.
- Repeat 16 rounds: $L_i = R_{i-1}$ and $R_i = L_{i-1}$ XOR $f(R_{i-1}, K_i)$.
- f function: expand R from 32 to 48 bits, XOR with the 48-bit round key, pass through 8 S-boxes (48 to 32 bits), then permute.
- Swap the halves and apply the Final Permutation to get the ciphertext.
- Round keys K1 to K16 are made from the 56-bit key by shifts and permutations. Decryption uses the same steps with keys in reverse order.

Today DES is weak because 56 bits can be brute-forced, so Triple DES or AES is used.